Compact proof certificates for ZFC consistency

Round 13 research report and reproducible equations

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Abstract

We give a smaller explicit direct proof-certificate construction for a Diophantine formulation of the consistency of ZFC. The main equation is a quartic in 1859 nonnegative integer variables; a distinct signed-quotient companion is a quartic in 3179 unrestricted integer variables. Their coefficients have absolute value below 2^{26} and are supplied in full. The main variable reduction is from the independently recounted pinned 3443-variable baseline, not an obsolete predecessor. The construction removes local beta remainder slacks, replaces copied bound circuits by fixed squaring chains, and makes canonical Pell witnesses point forward before eliminating their definitional coordinates. Its final synthesis composes these reductions with incoming eight-column projection and sign-orbit normalization, with those imported ideas credited explicitly.

A separate inherited native-code route now yields a twelve-integer equation of exact degree 2048 with 233 arithmetic operations. Its new quintic-carrier synthesis uses a supplied full-residue interpolation theorem. Additional complete equations have variable/degree pairs 13/1072, 14/378, 15/172, 15/170 and 16/98. Finally, four ten-integer norm circuits become shorter, while a leading-form argument proves degree 1080888 optimal only within the specified final-coordinate translation family. None of these results is a global minimum or a proof that ZFC is consistent. The report distinguishes ordinary mathematical derivations, inherited calibration, exact finite polynomial checks, and the end-to-end formalization still to be completed.

1 Results and the exact logical target

Fix the recursive ZFC axiom presentation and finite proof predicate used by the included calibration sources. Write

$$\mathrm{Con(ZFC)} \;:=\; \neg\exists p \in \mathbb{N}\ \mathrm{Ref}_{\mathrm{ZFC}}(p).$$

Here $\mathbb{N} = \{0, 1, 2, \ldots\}$ throughout. For every proposed polynomial one must specify its coefficient tuple and its domain; a count over $\mathbb{N}$ is not interchangeable with a count over $\mathbb{Z}$. The intended statement is a theorem of ZFC of the form

$$\mathrm{ZFC} \vdash \left(\neg\exists x \in D^n\ P(x) = 0\right) \;\longleftrightarrow\; \mathrm{Con(ZFC)}, \qquad D \in \{\mathbb{N}, \mathbb{Z}\}.$$

Equivalently, the construction must internally identify existence of a zero with a finite refutation. It must not assume consistency in proving that identification. Nor does the identification prove either side of the displayed equivalence. The finite identities and witness transformations given here are ordinary arithmetic arguments formalizable in ZFC. Their composition uses the named inherited refutation calibration. The package is not a new kernel-checked proof of the complete display.

The comparison snapshot is ProveIt commit 7e533c33819ee1f703b69f2778a767cdc069940d, authored at 03:32:10 UTC on 24 October 2026. It is a fixed snapshot, not an assertion that the repository has stopped changing.

For the other route, literal complete graph counts are as follows. A circuit operation is one live binary addition, subtraction or multiplication, including a square or scalar multiplication. Literal integer leaves and reuse are free. An abbreviation or circuit gate is not an additional unknown.

Construction	Domain	Variables	Degree	Terms	Height
Pinned direct baseline	$\mathbb{N}$	3443	4	15754	127302193
New direct variable focus	$\mathbb{N}$	1859	4	12275	63924066
New direct sparse option	$\mathbb{N}$	2560	4	11074	63923744
New direct integer companion	$\mathbb{Z}$	3179	4	23157	63923766

Table 1: Complete collected quartics. Height is the maximum absolute integer coefficient, after collecting equal monomials.

Family	Integer variables	Exact degree	Operations
Quintic five-root carrier	12	2048	233
Five-root coprime carrier	13	1072	192
Four-root coprime carrier	14	378	145
Three-root direct carrier	15	172	123
Three-root centered carrier	15	170	134
Two-root centered carrier	16	98	138
Ten-integer polar norm	10	1245324	467
Ten-integer linear translation	10	1179788	470
Ten-integer cubic translation	10	1114252	473
Ten-integer six-term translation	10	1080888	507

These rows cannot be ranked by a single unstated meaning of “simplest.” The direct route has small coefficient height and degree but many variables. The native route has very few variables and large fixed numerals. The centered fifteen-variable point trades eleven extra operations for two degrees relative to its direct sibling. We claim only the stated, audited tradeoffs.

1.1 What is new and what is inherited

The direct construction uses the numerical C222 proof calculus, its ZF-refutation calibration, and constructible-choice passage from ZF to ZFC refutations. It is independent of the K215/Jones/native coefficient route. The low-variable families retain the latter route’s coherent coefficient tuple and its ordinary calibration. Keeping the two chains separate is essential: a short evaluator identity cannot replace an unbounded compiler or Pell theorem.

The local work in this round consists of explicit source transformations, new carrier combinations, a credited projection and sign-normalization synthesis, polynomial identities, new exact circuits, and a restricted-family lower bound. The baseline comparison and all coefficient payloads are fixed before counts are made. Source files are retained with notices as inert mathematical provenance; no upstream executable is run by the release checks. The checkers supply finite implementation evidence, not a proof of consistency or a replacement for universal soundness and completeness.

1.2 How an explicit equation is specified here

For each direct equation the file `equation.json` gives an ordered list of variables and quadratic rows with literal integer coefficients. A row is encoded as a finite list (c, μ), where μ is a list of zero-based variable indices and denotes $c \prod_{i \in \mu} x_i$. The empty list denotes the constant monomial. The output is the sum of the squares of all rows. The accompanying `expanded_polynomial.json` gives its collected integer polynomial, not just a hash or a recipe for missing coefficients.

For a small-variable norm, the JSON specifies a finite acyclic graph with only literal or fixed-integer leaves and binary integer arithmetic. Every fixed-integer leaf has its actual byte payload in the archive, together with signedness, byte order and hashes. Hence these are ordinary fixed polynomials, not coefficient-parameter families. The factored displays below define the same equations more legibly than a huge monomial listing.

1.3 Comparison update from later incoming packages

After the fixed repository snapshot was reviewed, additional manuscripts and equation packets arrived. They change the comparison, but not the literal correctness of the equations in this report. They are not silently treated as files that were already present at the pinned commit.

A separate literal audit re-expanded the complete sums of squares in those direct packets and checked every supplied collected coefficient. Among them are unrestricted-integer quartics with the following metrics:

Variables	Quadratic rows	Collected terms	Height
3776	2015	20194	47959
3671	1942	26271	3835369234

The initial 4087-integer checkpoint was therefore not a new smallest integer count among the incoming packets. The final targeted synthesis in Section 2.10 instead gives 3179 unrestricted integers, fewer than both rows. It also improves the second row's monomial count and height, but not the first row's small coefficients or monomial count. The 1859-natural-variable construction is smaller in dimension than the later 2473-natural quartic, whose literal audit gives 2089 rows, 20580 terms and height 127808289. Projection and sign normalization from these incoming packages are credited prerequisites of that final synthesis, not claimed as independently new ideas here. These are domain-specific comparisons, not a total ordering.

A later thirteen-integer graph has 190 live binary operations and a claimed exact degree 1072, compared with 192 operations here. The operation count was independently verified; its cancellation-sensitive exact degree remains supported by its accompanying ordinary proof rather than that finite recount alone. Consequently the thirteen-variable row here is not presented as a current operation record. Two later twelve-integer graphs have 248 and 223 operations and reported degrees 4832 and 16000, respectively. Their syntactic degree upper bounds and operation counts were checked; exact attainment and the unbounded consistency calibration are separate authored claims. They provide further variable–degree tradeoffs rather than invalidating the smaller-degree thirteen- to sixteen-variable equations here.

The comparison addendum, audit receipts and payload identities are retained under `late_review/`. The original comparison archives and their separate checkers are not part of the compact offline suite; these receipts are comparison evidence, not a promise to reproduce omitted packages. The literal audit is not an independent reproof of all inherited calibrations or a claim that those later equations were derived in this round. We keep the principal $3443 \to 1859$ improvement explicitly tied to the stated fixed baseline.

2 Compressed direct proof-certificate quartics

The final selected direct construction gives a literal quartic in 1,859 nonnegative variables and a literal quartic in 3,179 unrestricted integer variables. The natural equation has 1,581 quadratic residuals, 12,275 monomials and coefficient height 63,924,066. The integer equation has 2,074

residuals, 23,157 monomials and height 63,923,766. Both heights are below 2^{26}. For its integer polynomial Q_8, the theorem is

$$\mathrm{Con(ZFC)} \quad \Longleftrightarrow \quad \neg\exists z \in \mathbb{Z}^{3179}\, Q_8(z) = 0.$$

This is a no-zero equivalence, not a proof that the polynomial has no zero, and not a new kernel-checked end-to-end translation.

We first establish the arithmetic and an eleven-column intermediate source, then project to eight columns and normalize auxiliary signs in Section 2.10. The reconstructible intermediate checkpoint has 2,170 natural variables or 4,087 integer variables; these are not the final selected counts. A retained sparse alternative with 2,560 natural variables and 11,074 monomials is also supplied. Its smaller monomial count is a genuine tradeoff against the selected 1,859-variable natural polynomial. The already improved baseline before this round has 3,443 natural variables, 2,838 residuals, 15,754 monomials and coefficient height 127,302,193. Late incoming integer comparisons and their advantages are stated at the end of this section.

The construction follows the direct numerical C222 proof calculus, independently of universal-polynomial or small-machine routes. Its calibration and indexed-Pell theorem are explicit inherited dependencies. All new transformations below are existence-preserving mathematical arguments, and the final files contain literal integer coefficients.

2.1 The exact direct source and the carrier argument

Write $I(x, y) = 2(x + y)(x + y + 1) + 4y$, and define

$$A(x, y) = I(x, y) + 2, \quad M(x, y) = I(x, y) + 1, \quad E(x, y) = I(x, y) + 3, \quad \neg x = I(x, 1).$$

These are numerical constructors, not arithmetic assertions about sets. The four schema coordinates a, b, c, d and twelve padded Horn branches are exactly the pinned numerical C222 source. The target is code 1 and the free truth code is 0. The explicit axiom constructors, including the bundled axiom and its disequality guard, occur literally in `build_compact.py`; the inherited sparse source and calibration modules are retained in `sources/`.

The dependency is the C222-to-ZF refutation theorem followed by the constructible-choice ZF-to-ZFC refutation theorem, not execution of a small universal machine. In particular, no assumption that every natural number decodes a well-formed formula is silently added.

For $N, m \geq 1$, let $M = (m + 1)^N$ and bound each rule label p_i, q_i, v_i by m. The accumulator source is

$$S_0 = M + 1, \quad S_N = 1, \qquad S_{i+1}(1 + v_i M) = S_i(1 + p_i M)(1 + q_i M), \quad i < N. \tag{1}$$

Every row also requires the pinned Horn rule $(p_i, q_i) \to v_i$. Telescoping gives an equality at M between

$$F(X) = \prod_{i<N} (1 + v_i X), \qquad G(X) = (1 + X) \prod_{i<N} (1 + p_i X)(1 + q_i X).$$

If $\deg G \geq N + 1$, nonnegative coefficients give $G(M) \geq M^{N+1}$, whereas

$$F(M) \leq (1 + mM)^N < ((m + 1)M)^N = M^{N+1}.$$

Thus both products have at most N nonconstant factors. Their coefficient sums at 1 are at most $(m + 1)^N = M$, and every individual coefficient is strictly below M: a nonconstant product has its positive constant term and another positive coefficient, while the constant product is $1 < M$. A nonzero polynomial $F - G$ cannot vanish at M, since its least nonzero coefficient would be a nonzero multiple of M with absolute value below M. Hence $F = G$.

The distinct factors $1 + jX$, $j > 0$, now recover the exact nonnegative Horn flow balance for target 1. A finite single-conclusion flow implies derivability: if the target were not derivable, sum the flow coordinates outside the deductive closure. Each valid rule contributes at most zero, but the target contributes one. Conversely, expand a finite proof tree from the open obligation 1 and let S_i be the product of the weights of all open obligations. This gives (1). Truth padding has weight one. The proof uses single conclusions and nonnegative multiplicities; it does not justify arbitrary cyclic proofs or signed flows.

2.2 Eleven coordinates instead of thirteen

Use the beta sequence function

$$\beta(x, y, i) = x \bmod (1 + (i + 1)y).$$

For every finite natural sequence, its scale y may be chosen strictly larger than every sequence entry and divisible by $N!$. The moduli for $i = 0, \ldots, N$ are then pairwise coprime: a common divisor is coprime to y and divides a positive index difference at most N, which divides y. The Chinese remainder theorem supplies x.

Parameterize

$$N = N_0 + 1, \quad m = m_0 + 1, \quad M = M_0 + 1 = (m + 1)^N, \qquad y = M_0 + 2 + y_0,$$

with all displayed underlying parameters natural. The two endpoint rows are simply

$$x = M_0 + 2 + (y + 1)u_0, \qquad x = 1 + \big(1 + (N + 1)y\big)u_N. \tag{2}$$

They are canonical remainder statements because $M_0 + 2 \le y < y + 1$ and $1 < 1 + (N + 1)y$. The parameterization alone permits $S_0 = y$; completeness chooses $y > S_0$, and the first bounded row also enforces this strict inequality. No incorrect strict endpoint inequality is assumed.

At row i, retain only

$$w = (p, q, v, a, b, c, d, S, S', u, u') \in \mathbb{N}^{11}.$$

Besides the Horn condition and the accumulator row, require

$$x = S + \big(1 + (i + 1)y\big)u, \qquad x = S' + \big(1 + (i + 2)y\big)u'. \tag{3}$$

The CRT compiler bounds p, q, v by m, bounds S, S' by $y - 1$, and bounds a, b, c, d, u, u' by one independent U. Therefore (3) already forces the canonical beta digits. The two local remainder slack coordinates and their two binomial range certificates disappear.

For completeness first choose the proof, N, m, M, and its finite accumulator sequence, then choose y strictly above every state, then choose x by CRT, and finally choose U above the finitely many schema and quotient coordinates. None of these choices depends on the later bounded-universal CRT encoding. This order excludes circular bounds.

2.3 Factoring the common Horn conditions

Let $f_1, \ldots, f_8$ be the eight unconditional axiom codes, let b_{bundle} denote the bundled axiom code, and let

$$g = ((c - a)(c - b))^2 - (d + 1)^2$$

be its guard residual. Here b in the guard remains the schema coordinate. Put

$$J = v \prod_{r=1}^{8} (v - f_r), \qquad K = (v - b_{\text{bundle}})^2 + g^2.$$

The following fixed nonnegative polynomial has exactly the pinned Horn zero set:

$$H_{\mathrm{Horn}} = (q - I(p,v))^2 \left[q^2 + (v - A(a,p))^2(p^2 + J^2)(p^2 + K)\right]. \tag{4}$$

Indeed its first factor is modus ponens. Otherwise the bracket vanishes exactly when $q = 0$ and at least one of these alternatives holds: $v = A(a,p)$; or $p = 0$ and $J = 0$; or $p = 0$ and $K = 0$. These are generalization, truth or one of the eight unconditional axioms, and the guarded bundled axiom, respectively. Every factor used in this argument is nonnegative. This is an equivalence of zero sets, not an asserted polynomial identity with the old twelve-factor source.

Let P be (4) plus the squares of the accumulator residual and the two residuals in (3). The signed arithmetic circuit is split as $P = E - F$, where E, F have nonnegative coefficients, using

$$(e,f) + (g,h) = (e+g, f+h), \qquad (e,f)(g,h) = (eg + fh, eh + fg),$$

and $(e-f)^2 = (e^2 + f^2) - 2ef$. Thus $E = F$ holds exactly when every required local condition holds.

2.4 One small circuit for a very large majorant

It is unnecessary to evaluate a second full copy of the source circuit at all coordinate upper bounds. Put

$$R = 4 + N + x + y + m + M_0 + U.$$

Every input at which the upper source is evaluated is at most R: the index is N, the label bounds are m, the six other private bounds are U, and y is an upper bound for each state (slightly larger than its actual bound $y - 1$). The external inputs are also included in R.

A structural exponent certificate bounds every nonnegative circuit node by a power of R. An input receives exponent one. A constant c receives the least e with $c \le 4^e$. At addition take $\max(e_1, e_2) + 1$, since $R \ge 4 \ge 2$; at multiplication take $e_1 + e_2$. Induction on the circuit proves the bound at every node. For the actual factored source the root exponents are 6,893,154 and 6,893,148. Rounding upwards to $2^{23} = 8,388,608$ gives

$$H = R^{2^{23}} \ge N, \qquad H \ge E(N,z,u), F(N,z,u). \tag{5}$$

Here the upper state entries in u may be taken to be y, as just noted. The literal compiler computes H by exactly 23 successive squarings. There is no variable exponent or power-representation macro at this step. The larger numerical value of the bound has essentially no circuit cost. The exported exponent certificate records every abstract circuit node and every one of these 23 squarings.

2.5 One bounded-quantifier certificate with direct cancellation

Set

$$T = H!, \quad D_i = 1 + (i+1)T, \quad B = \prod_{i<N} D_i.$$

For actual coordinate bounds $u = (m, m, m, U, U, U, U, y-1, y-1, U, U)$, the bounded row family is equivalent to existence of positive encoded columns A_j and a natural encoded index C with

$$B \mid \binom{A_j}{u_j + 1}, \qquad B \mid 1 + (C+1)T, \qquad E(C,z,A) \equiv F(C,z,A) \pmod{B}. \tag{6}$$

In particular, the two state binomial lower arguments are exactly y; no subtraction is needed in the finite arithmetic circuit.

For soundness fix $i < N$ and take a prime $p \mid D_i$. It is coprime to $T = H!$, hence $p > H$. The identity

$$(u_j + 1)!\binom{A_j}{u_j + 1} = \prod_{r=0}^{u_j}(A_j - r)$$

implies that $A_j \equiv w_j \pmod p$ for some $0 \le w_j \le u_j$. This remains true when $A_j \le u_j$, as one factor is zero. The index condition gives $C \equiv i \pmod p$. The evaluation congruence then holds for $E(i, z, w), F(i, z, w)$; both lie in $[0, H]$, so they are equal.

For completeness the D_i are pairwise coprime because $H \ge N$. The radius bound supplies more than needed for soundness: $H \ge R$ and $R \ge u_j + 1$ for every coordinate. Consequently $L_j = (u_j + 1)!$ divides $H! = T$, and $\gcd(B, L_j) = 1$. Choose ordinary CRT representatives $A_j \equiv w_j^{(i)} \pmod{D_i}$ and $C \equiv i \pmod{D_i}$. The falling-factorial numerator then vanishes modulo every D_i and hence modulo B. Cancelling L_j, which is coprime to B, gives $B \mid \binom{A_j}{u_j+1}$. Adding a sufficiently large multiple of B makes $A_j > 0$, preserving the falling-product congruence and hence, by the same cancellation, the binomial divisibility. The valid rows and polynomial congruence give the evaluation condition in (6).

This final radius construction therefore needs neither prime-supported CRT lifts nor their proof-only products. Those are necessary in the more general current-baseline theorem without a factorial-size guard, but the very cheap radius bound restores the simpler direct cancellation proof. There is no circularity: all coordinate bounds and H are chosen before T, B and the encoded columns.

2.6 All arithmetic operators are eliminated

The exact arithmetic identities are

$$Q_0 = (1 + (N + 1)T)^N + 2, \qquad B = \left(T^N N!\binom{Q_0 + N}{N}\right) \bmod (TQ_0 - 1), \tag{7}$$

$$\binom{n}{k} = \left\lfloor \frac{(2^n + 2)^n}{(2^n + 1)^k} \right\rfloor \bmod (2^n + 1), \tag{8}$$

$$r = (n + 2)^{n+2}, \quad C_0 = \binom{r + n}{n}, \qquad n! = \left\lfloor \frac{(r + 1)^n + C_0 - 1}{C_0} \right\rfloor. \tag{9}$$

For (7), reduction of the numerator modulo $TQ_0 - 1$ gives $\prod_{j=1}^{N}(1 + jT)$, and the chosen Q_0 makes this product strictly smaller than the modulus. Division uses $a = dq + r$ and $d = r + s + 1$. The circuit for (8) combines its two divisions using one high-digit coordinate. Formula (9) uses the equivalent equation $(r + 1)^n + C_0 = C_0 n! + r_0 + 1$, with $0 \le r_0 < C_0$. The included factorial and bounded-forall representation sources state these identities with their boundary cases.

There are exactly 49 positive-power graph calls, 14 binomial calls, and two factorial calls. Positivity is established before each call: $N, m \ge 1$; $H \ge 4$; encoded columns are parameterized as $A_j = \alpha_j + 1$; range lower arguments are $m + 1$, $U + 1$, or $y \ge 2$; all auxiliary factorial/binomial upper arguments and power bases are positive. Thus deleting zero-base and zero-exponent cases is licensed for this construction, not for the unrestricted exponentiation relation.

2.7 The parametric positive-power graph

For positive n, k, let m be the proposed output. There are twelve natural certificate coordinates

$$a_2, y_k, g, d, X, U, q_4, q_B, q_S, q_R, q_P, Z.$$

Define, rather than allocate additional unconstrained coordinates,

$$W = n + k, \qquad A = a_2 + 2, \qquad Y = k + y_k, \qquad T_0 = m + g + 1,$$
$$V = Y^2(d+1), \qquad B_0 = 1 + 4Y(q_4 + 1), \qquad S = X + Uq_S, \qquad R_0 = k + 4Yq_R.$$

Writing $\mathcal{N}(a, x, y)$ for $x^2 + y^2 - a^2y^2 = 1$, require precisely seven polynomial conditions:

$$\mathcal{N}(A, X, Y), \quad \mathcal{N}(A, U, V), \quad \mathcal{N}(B_0, S, R_0),$$
$$B_0 = A + Uq_B, \qquad X + Yn = YA + m + T_0q_P, \qquad 2An = T_0 + n^2 + 1,$$
$$\mathcal{N}(W + 1, A, WZ). \tag{10}$$

These are circuit expressions; flattening their products gives quadratic residuals. The explicitly assigned B_0 is strictly greater than one, including when $q_4 = 0$. There is no missing bound on this Pell parameter.

Soundness. The definitions give $A > 1$, $Y \geq k > 0$, $V > 0$, $Y^2 \mid V$, and $B_0 > 1$. The three forward congruence equations imply the three modular conditions of the pinned indexed-Pell criterion. Thus (X, Y) is the kth Pell solution for parameter A. The last norm gives

$$A^2 = 1 + W^3(W + 2)Z^2.$$

As $A \geq 2$ and $W \geq 1$, necessarily $Z \geq 1$ and $A > W \geq n$. Therefore the rearranged power congruence uses the same ordinary subtraction as the pinned power theorem. Also $m < T_0$, and the deleted common-bound slacks can be restored as $W - n = k$ and $W - k = n$. The positive-power criterion yields $m = n^k$.

Completeness and the direction of every quotient. Let $(X_a(j), Y_a(j))$ denote the natural Pell sequence. At $W = n + k$ choose

$$A = X_{W+1}(W), \quad WZ = Y_{W+1}(W), \quad T_0 = 2An - n^2 - 1, \quad X = X_A(k),\ Y = Y_A(k).$$

The standard congruence $Y_{W+1}(W) \equiv W \pmod{W}$ supplies Z. The usual fixed-bound power construction gives $n^k < T_0$ and the power congruence. For its direction set $F_j = X_A(j) - (A - n)Y_A(j)$. Then $F_0 = 1, F_1 = n, F_{j+1} = 2AF_j - F_{j-1}$. Inductively $F_j \geq F_{j-1} \geq 0$ and $F_{j+1} \geq (2A - 1)F_j \geq nF_j$, so $F_k \geq n^k$. Thus $q_P = (F_k - n^k)/T_0$ is natural.

In the explicit completeness proof of the indexed-Pell criterion set $h = 2kY$, $U = X_A(h)$, $V = Y_A(h)$, and choose B_0 by CRT with $B_0 \equiv A \pmod{U}$ and $B_0 \equiv 1 \pmod{4Y}$. That proof has $A < U$, so its natural CRT representative satisfies $B_0 \geq A$. Take $S = X_{B_0}(k)$ and $R_0 = Y_{B_0}(k)$, at the same index k. The Pell sequences are monotone in the parameter, hence $S \geq X$; also $R_0 \geq k$. Consequently all q_B, q_S, q_R are natural. Because $B_0 > 1$ and $Y > 0$, its quotient in $B_0 = 1 + 4Yq$ satisfies $q \geq 1$, so write $q = q_4 + 1$. The remaining slacks follow from the displayed inequalities and divisibility. This proves completeness of (10), not merely its one-way soundness. The separate orientation supplement records the monotonicity proof and exact pinned source lines.

2.8 Quartic flattening, sparse alternative, and integer domain

Every nonnegative arithmetic addition or multiplication receives a triangular defining row. Nonnegative affine gate values may be substituted without raising residual degree; the implementation substitutes only expressions of support at most four, a size tradeoff rather than a semantic restriction. The complete substitution map is supplied. The final polynomial is the sum of

squares of all remaining quadratic rows. Over either $\mathbb{N}$ or $\mathbb{Z}$, a sum of integer squares vanishes exactly when all rows vanish. Exact expansion and collection give the reported quartics.

The 2,560-variable sparse alternative retains semantic names for the Pell outputs rather than making all of (10)'s definitions affine substitutions. It has fewer collected monomials. Its orientation, short-beta source, CRT block and radius bound are otherwise the same. The intermediate shared-offset construction is also justified: finitely many signed quotients can be written $v_i - h$ with one common natural h, chosen after all other witnesses. It is superseded by the forward form for the final natural variants, except that one final CRT congruence still uses two natural quotients.

For the 4,087-variable integer companion, congruence quotients themselves need not be nonnegative. There are exactly 197 unrestricted quotient coordinates: four per positive-power graph and one final CRT quotient. In (10), S, R_0 are retained as separate natural semantic coordinates, with their defining equalities, while q_S, q_R may be signed. The other parametric definitions are unchanged. In particular $B_0 > 1$ still follows from $B_0 = 1 + 4Y(q_4 + 1)$ with natural q_4; it is not incorrectly inferred from signed q_B. Soundness reads congruences directly and applies the same indexed-Pell and power theorems. Completeness uses the natural witness construction already proved. The final CRT quotient may be signed because (6) is a divisibility condition.

Only 607 further nonnegativity assertions are necessary. An exact sign certificate infers the other 44 of the 651 nonquotient semantic coordinates from polynomial equalities. Each inference is of the form $cx = P$ with $c > 0$ and P having nonnegative coefficients; every variable occurring to an odd exponent in a monomial of P must already be known nonnegative. Even powers need no sign premise. The supplied application order is an acyclic proof of all required signs. Signed quotient gates are not asserted to be nonnegative.

For each remaining seed x, append the quadratic residual

$$2x - a(a+1) - b(b+1) - c(c+1) = 0. \tag{11}$$

For arbitrary integers a, b, c, each consecutive product is nonnegative, so (11) forces $x \geq 0$. Conversely every natural x is a sum of three triangular numbers by Gauss's theorem. Equivalently, apply Legendre's three-square theorem to $8x + 3$: the three squares must all be odd modulo 8, and subtracting their three units yields (11). The 607 seeds add 1,821 integer coordinates and 607 quadratic rows to the 2,266-coordinate mixed-domain template. This proves the counts $2266 + 1821 = 4087$ and $1924 + 607 = 2531$, and preserves degree four.

2.9 Source pins, reproduction, and the claim boundary

The calibration files are pinned to `VladimirReshetnikov/ProveIt` commit `08200eb10fde2d b408c69dffd840d6920a569b1f`; the comparison source is the fixed-power-bound candidate at commit `7e533c33819ee1f703b69f2778a767cdc069940d`. The exact theorem names, file paths and SHA256 values are listed in `SOURCE_PINS.json` and the inherited calibration inventory. The all-forward Pell proof additionally pins the official Mathlib file at commit `4dc43842a11f257ffba2a611c943a133a65cb4b2`, SHA256 `5c5756cdf2fa70fb77af9daa9c7167 c65749ac616bde8f5fdfeed2cb958b3ed5`. It is inspected as source text, not executed.

The builders are newly authored standard-library Python implementations. No upstream executable was imported or run. Their baseline mode independently reconstructs all five published numerical baseline metrics. The final release includes the exact sparse equations, every residual, raw circuits and macro bindings, affine restoration maps, structural radius certificates, integer sign certificates, source pins, and independent finite-audit receipts. Reproduction commands are in `README.md` and the file whitelist is machine-readable.

These finite checks establish literal polynomial and certificate facts, not consistency of ZFC and not the universal mathematical dependencies. The logical chain is: pinned numerical refutation calibration; finite single-conclusion proof flow and the carrier lemma; short-beta source; factored Horn zero set; bounded CRT with factorial cancellation; radius bound; explicit arithmetic identities and Pell graphs; quartic flattening; and, for unrestricted integers, sign closure and the three-triangular-number theorem. No nonpolynomial operation remains in the final equation.

2.10 Final source projection: eight columns and no private bound

The source projection in this subsection is explicitly credited to the incoming *ConZFC Direct Projection 7617* package, whose preprocessing proof is preserved as `sources/INCOMING_PROJECTION_PROOF.md`. The new result here is its composition with the sharper carrier, parametric positive-power graphs, dyadic majorants, quotient bound x, and sign-normalized integer compilation. We do not claim that eliminating these three label columns was first discovered in this construction.

Each rule can be parameterized by just a, b, c, d:

rule	p	q	v
truth	0	0	0
eight ordinary axioms	0	0	$f_j(a,b,c,d)$
bundled axiom	0	0	$f_{\text{bundle}}(a,b,c,d)$
modus ponens	a	$I(a,b)$	b
generalization	b	0	$A(a,b)$

The bundled row additionally requires $((c-a)(c-b))^2 = (d+1)^2$. No other row has a guard. These substitutions are exactly the pinned C222 rules, including the independent parameter and premise of generalization.

For a chosen branch j, substitute its labels (p_j, q_j, v_j) into the accumulator residual

$$L_j = S'(1+v_jM) - S(1+p_jM)(1+q_jM).$$

Let $C_j = L_j^2$ except at the bundle, where $C_j = L_j^2 + g^2$. The new local polynomial is

$$P_8 = \prod_{j=0}^{11} C_j + \big(x - S - (1+(i+1)y)u\big)^2 + \big(x - S' - (1+(i+2)y)u'\big)^2. \tag{12}$$

Its zero set is exactly one valid branch-specific update together with the two common beta reconstructions. As all branch factors are nonnegative, the product implements the finite disjunction without selecting a rule by an extra coordinate. This is existential projection of the three deterministic labels, not an assertion that the polynomial in (12) is algebraically identical to the eleven-coordinate source.

Padding chooses the parameter bound before the carrier. Given a finite proof-tree expansion, choose $N \geq 1$ at least its length and at least every schema parameter in the expansion. For modus ponens, this may include a very large formula code; no practical bound on the resulting N is claimed. Pad to exactly N rows using truth, whose labels are $(0,0,0)$ and whose update leaves S unchanged. Thus all four parameters may be bounded by N without changing refutability.

Every branch label is a nonnegative-coefficient polynomial in those four parameters. Apply the structural exponent certificate to its literal constructor DAG with radius $N+4$. The largest certified exponent is 3,445,903, so the fixed polynomial

$$m = (N+4)^{2^{22}} \tag{13}$$

bounds every label. It is computed by exactly 22 squarings, not by a variable-exponent graph and not by a duplicate evaluation of the bundled axiom. It satisfies $m \geq 1$. Choose $M = (m+1)^N$ and $M_0 = M - 1$. The carrier proof already given applies verbatim to these labels. The choice order is proof, padded N, then m, M, then the accumulator sequence, then y, x. There is no self-referential bound.

The eight local coordinates are

$$(a, b, c, d, S, S', u, u'), \qquad \text{bounds } (N, N, N, N, y-1, y-1, x, x). \tag{14}$$

The use of x for the two quotient bounds is an additional simplification: canonical beta reconstructions have $x = S + Du$ with $S \geq 0, D \geq 1$, hence $u \leq x$, and similarly for u'. No new private bound U is needed. Conversely, the state bounds in (14) make those reconstructions canonical, just as in the eleven-column proof. The endpoints and parameterization $y = M_0 + 2 + y_0$ are unchanged. The four outer natural coordinates are N_0, x, y_0, M_0, with $N = N_0 + 1$.

A second dyadic radius proves the shared CRT bound. Split the actual signed circuit for (12) as $P_8 = E_8 - F_8$ and put

$$R = 4 + N + x + y + M_0, \qquad H = R^{2^{23}}. \tag{15}$$

All inputs at the upper substitution of (14) are at most R; for the two states we safely use y instead of $y - 1$. The two root exponents from the structural certificate are both 6,893,248, below 2^{23}. Therefore H bounds both source evaluations and N. It also dominates every binomial cutoff $N + 1, y, x + 1$. The same CRT theorem, with direct factorial cancellation, applies. Only 23 successive squarings compute (15). Notice that m need not occur in this source radius: after label projection the local source computes its labels from the four bounded parameters and uses only M_0 as its carrier input.

The final eight-column arithmetic uses exactly 40 positive-power graphs, 11 binomial graphs and two factorial graphs. The count is transparent: 24 powers for the eight coordinate binomials, five for each of the two factorials, five other powers in the modulus product, and one carrier power. In this accounting the five other product powers are its two explicit powers and the three powers of its extra binomial; the factorial at N has already been counted. No arithmetic macro, variable-length product or variable exponent remains after flattening.

2.11 Existential sign normalization for the final integer equation

The incoming signed-quartic package supplies the sign-orbit method and its two pivots per positive-power graph; its proof is preserved as `sources/INCOMING_SIGN_NORMALIZATION_PROOF.md`. The final application here is checked against the new projected parametric system, not assumed merely because it worked for a predecessor.

For a family of residuals r_j, a diagonal sign action $x_i \mapsto (-1)^{\epsilon_i} x_i$ preserves the common zero set whenever all monomials of each row have equal parity against ϵ. Then r_j is changed to either r_j or $-r_j$. For each of the 40 power calls there are two independent actions:

- Flip its auxiliary Z and the product wire WZ. The only other use of that wire is through its square in the last norm.
- Flip its Pell coordinate U and its two unrestricted quotients q_B, q_S. The products Uq_B, Uq_S are unchanged, while the norm uses U only through U^2.

The actions leave every other semantic coordinate unchanged. They commute, and do not change any retained sign guard. At any integer zero, apply a pivot's action exactly when that pivot

is negative. This produces another zero in which all 80 pivots are nonnegative. This is an existence-preserving normalization, not a claim that the original zero already had those signs.

The released normalization certificate specifies all flipped raw and exported variables and the parity of every residual under each action. The separate independent checker verifies these finite identities; the argument above explains their universal meaning. Gate values involving signed quotients remain allowed to be negative. No signs of X, S, R_0 are discarded by an unsupported symmetry argument.

The mixed-domain template has 1,937 variables and 1,660 quadratic rows. It has 161 unrestricted congruence quotients: four per power call and one final CRT quotient. There are 529 other semantic coordinates. Before normalization, exact polynomial sign closure requires 494 seeds and infers 35 more signs. The 80 pivots are among these seeds and can instead be normalized. Thus only

$$414 + 80 + 35 = 529$$

semantic signs need to be guarded, normalized, or inferred, respectively. The 414 three-triangular-number certificates add $3 \cdot 414 = 1242$ integer variables and 414 quadratic rows. Consequently the final integer counts are

$$1937 + 1242 = 3179, \qquad 1660 + 414 = 2074.$$

The resulting ordinary quartic Q_8 has 23,157 monomials and coefficient height 63,923,766. The natural companion has 1,859 variables, 1,581 rows, 12,275 monomials and height 63,924,066. Both heights are below 2^{26}. The literal files are `projected/expanded_polynomial.json` and `projected_integer/expanded_polynomial.json`.

Complete equivalence and comparisons. A zero of the integer polynomial first yields an integer zero of the mixed template with the 414 guarded signs. Normalize the 80 pivots and replay the 35 sign implications. All nonquotient arithmetic inputs are then natural. The parametric Pell certificates establish every required power; formulas (7)–(9) establish every other arithmetic macro; the CRT certificate recovers all eight-column rows; their parameter and label bounds license the sharp carrier and hence a C222 refutation. The pinned calibration gives a ZFC refutation. Conversely a finite refutation gives the padded source in (12)–(14), all arithmetic certificates, natural power witnesses, and the 414 triangular representations, hence an integer zero. Therefore

$$\mathrm{Con(ZFC)} \quad \Longleftrightarrow \quad \neg\exists z \in \mathbb{Z}^{3179}\, Q_8(z) = 0.$$

All steps are ordinary mathematics formalizable in ZFC. Finite artifact checks do not replace the stated unbounded arithmetic and calibration theorems, and no new proof-assistant certification or proof of consistency is claimed.

The late incoming direct integer alternatives have 3,776 variables, 20,194 monomials and height 47,959; and 3,671 variables, 26,271 monomials and height 3,835,369,234. The new 3,179-variable equation improves the latter on variables, monomials and coefficient height, but does not improve the former's monomial count or its much smaller coefficients. It should not be described as dominating every current direct equation. The 2,560-natural-variable sparse companion likewise remains relevant because its 11,074 monomials are fewer than the projected natural companion's 12,275. These are explicit tradeoffs, not an absolute minimality claim.

3 Measured alternative certificate architectures

3.1 Negative CRT moduli as a valid alternative

Suppose the row condition is $\prod_j C_j = 0$ and $R_0 = 0$, where all C_j, R_0 are sums of squares. Bound each C_j and R_0 separately by K, with $K \geq 4, N$ and all range cutoffs. Put

$$T = K!, \quad D_i = (i+1)T - 1, \quad B = \prod_{i<N} D_i.$$

The D_i are pairwise coprime by the usual difference argument, and each is $3 \pmod 4$. Hence each has a prime divisor $p \equiv 3 \pmod 4$; since $p \nmid K!$, necessarily $p > K$. Using the usual binomial range certificates, the index congruence $B \mid (C+1)T - 1$, and

$$\left(\prod_j C_j(C, A)\right)^2 + R_0(C, A)^2 \equiv 0 \pmod B$$

recovers every row. Indeed, over such a prime, a sum of two squares is zero only when both terms are zero. Primality then selects a vanishing C_j, and the separate bounds turn the selected branch and R_0 into exact zeros. Ordinary CRT proves completeness.

No variable-length product remains in the output. With $Q = (1 + (N+1)T)^N + 2$ one has

$$B = \left(T^N N! \binom{Q+N}{N}\right) \bmod (TQ+1).$$

The numerator is $\prod_{j=1}^{N}(TQ + jT)$, which reduces to $\prod_{j=1}^{N}(jT - 1)$, and $B < Q < TQ + 1$. For $N = 0$ the product is 1, the congruences modulo 1 are vacuous, and the same identity holds. The rounded-radius negative-modulus benchmark has 2678 natural variables, six more than its matched positive-modulus comparison. It is a proved alternative, not the recommended minimum.

3.2 Factorial separation without a coefficient height

If $M \geq 1$ and $\max(m-1, 0)! \mid M$, then the numbers $1 + jM$, $1 \leq j \leq m$, are pairwise coprime. A common divisor for labels $j < k$ is coprime to M and divides $k - j$, while $k - j$ divides M. Therefore it is 1. Prime valuations now recover every positive label multiplicity from a product equality, for arbitrary proof length. The factors themselves need not be prime. Label zero contributes the unit factor and remains invisible. For $m = 0$ the target factor $1 + M > 1$ makes the equality impossible; for $m = 1$ any positive M works.

Thus $M = m!$ is a correct replacement carrier when $m \geq 1$. It is chosen before the obligation stream, never from its later CRT bound. The matched native-graph benchmark costs 2872 instead of 2672 variables: one extra factorial, one extra binomial and four extra power graphs. A completely fixed M without a label restriction is unsound, as shown by $1 + (M+2)M = (1+M)^2$ and the underivable single rule $(1, 0) \to M + 2$.

All arguments above are finite arithmetic theorems formalizable in ZFC. They remain internal theorems in nonstandard models; they do not rely on external standardness of a proposed certificate. The included finite tests and literal polynomial replays verify implementation fidelity, not a new kernel proof of those all-input statements.

4 Coprime scale carriers and fixed prime separation

A low-degree scale and fixed prime separation can be combined without adding a variable. The resulting complete integer equations are

Construction	Variables	Exact degree	Binary operations
Normalized five-root norm	13	1072	192
Ordinary four-root norm	14	378	145
Ordinary three-root norm	15	172	123
Centered three-root norm	15	170	134
Centered two-root norm	16	98	138

All variables range over unrestricted integers. These are explicit cost–degree tradeoffs, not global minima. In particular, the first row improves the incoming 13/1088/195 point; the second substantially lowers both degree and evaluation cost from the inspected 14/652/163 point. The 16-variable row lowers degree but increases cost relative to 16/108/132.

The source is the literal signed CRT parent `degree88_17.json`, with its inherited ordinary native compiler and K215 calibration. We restore each of its five residuals below. This is not a new Lean/Rocq certificate, and it proves neither consistency nor a historical or global optimality claim.

4.1 Complete common prefix

The base thirteen coordinates are

$$j, C, z, h, r, v, w, T, c, n, t, p_2, p_3.$$

The fixed six numerals are the coherent source-34 tuple β, K_0, M_C, M_F, with M_F already shifted, and $A_{\mathrm{cone}} = 2^{1113}$, $B_{\mathrm{cone}} = 2^{666104}$. In particular $\beta = 2^{1064921} - 1$. All named registers below are polynomial abbreviations, not extra quantified variables:

$$\begin{aligned}
J &= 6j + 1, & q &= \beta J + 1, & W &= 12w + 4,\\
X &= qW, & Y &= qv, & E &= XY,\\
a &= E + Y, & H &= 4a + 3, & \Delta &= a^2 + H,\\
F &= (K_0 + W)C - z(q-1), & & & &\\
R &= (q(q-F) - C)(q^2-1) + (M_C + qM_F)J, & & & &\\
k &= R + 1 + hE, & \eta &= c - Yk, & \zeta &= k - \eta,\\
u &= EYk, & A_\tau &= u(u+k) + 1, & A_m &= 1 + \Delta c^2,\\
b_0 &= X + ac, & I &= (2r+1)c^2, & M &= 2q^2,\\
U &= MH, & e &= v - q^2. & &
\end{aligned}$$

The determinant and guard radicand are

$$\begin{aligned}
d &= (A_{\mathrm{cone}}F)(B_{\mathrm{cone}}C) - (q - A_{\mathrm{cone}}F - B_{\mathrm{cone}}C)^2,\\
A_g &= d\eta\zeta - (t^2 - d - \eta\zeta + 1)^2 - 1 - p_2^2 - p_3(p_3+1).
\end{aligned}$$

For the ordinary variants, add the coordinate f and set

$$Q = I^2, \qquad S = Q - \Delta(f^2 - 1), \qquad V = fcT - f^2R - c.$$

For the thirteen-variable normalized variant instead use

$$A_f = 1 + \Delta I^2, \qquad Q = \Delta^2 I^2, \qquad V = A_f(cT - R) - c.$$

In both cases put $\delta = Q - 1$, $C_y = QV^2 - 1$. The rational expression $A_y = C_y/\delta$ is proof notation only, never a circuit division. Where the coordinate γ is retained, write $D = (b_0 + H\gamma)^2 - A_m$. Where p_1 is retained, write $G = A_g - p_1^2$.

4.2 Five literal norm specifications

Let

$$\mathcal{F}_m(x; a_1, \ldots, a_m) = \prod_{\varepsilon \in \{-1,1\}^m} \left(x + \sum_i \varepsilon_i \sqrt{a_i} \right).$$

Sign invariance proves that this is an integer polynomial. The following expressions are cleared symbolically; their JSON circuits use only binary addition, subtraction and multiplication.

For thirteen variables take $Z = U(n + V) + Mb_0 + He$ and

$$N_{13} = \delta^{16} \mathcal{F}_5 \left(Z; 81U^2 A_f, 9U^2 A_\tau, M^2 A_m, 729U^2 A_g, \frac{U^2 C_y}{\delta} \right).$$

The equation is $N_{13} = 0$.

For fourteen variables, add f, retain the same Z, and use

$$N_{14} = \delta^8 \mathcal{F}_4 \left(Z; 9U^2 A_\tau, 81U^2 A_g, M^2 A_m, \frac{U^2 C_y}{\delta} \right), \qquad P_{14} = (1 + N_{14})(1 + S) - 1.$$

For fifteen variables, add f, γ, and use

$$N_{15} = \delta^4 \mathcal{F}_3 \left(M(n + V) + e; 9M^2 A_\tau, 81M^2 A_g, \frac{M^2 C_y}{\delta} \right), \qquad P_{15} = (1 + N_{15})(1 + S)(1 + D) - 1.$$

The centered variants use $t_0 = 2u + k$, $\ell = k^2 - 4$. For fifteen variables add f, p_1 and set

$$\begin{aligned} Z_c &= U(n + 3t_0 + 2V) + 2Mb_0 + He, \\ N_{15c} &= \delta^4 \mathcal{F}_3 \left(Z_c; 36U^2 A_\tau, 4M^2 A_m, \frac{4U^2 C_y}{\delta} \right), \\ P_{15c} &= (1 + N_{15c})(1 + S)(1 + 4G) - 1. \end{aligned}$$

For sixteen variables add f, γ, p_1, put $x = Mn + e$, and define

$$\begin{aligned} j_0 &= \delta\{x^2 + M(6t_0 + 4V)x + 9M^2\ell\} - 4M^2(V^2 - 1), \\ h_0 &= 12\delta t_0 V M^2, \\ N_{16} &= j_0(j_0 + 2h_0) + 144\delta M^4\{4A_\tau(1 - V^2) + \delta\ell V^2\}, \\ P_{16} &= (1 + N_{16})(1 + S)(1 + D)(1 + 4G) - 1. \end{aligned}$$

The exact identity, valid also at $\delta = 0$, is

$$N_{16} = \delta^2 \mathcal{F}_2 \left(M(n + 3t_0 + 2V) + e; 36M^2 A_\tau, \frac{4M^2 C_y}{\delta} \right)$$

after symbolic clearing. An independent integer-coefficient check compares all 29 nonzero terms in the centered identity.

4.3 Unit products recover the rows before denominator use

For integers, if residuals R_i never equal -2, then

$$(1+N)\prod_i (1+R_i) = 1 \quad \Longleftrightarrow \quad N = R_1 = \cdots = R_s = 0.$$

Indeed every factor of the integer unit 1 is ± 1. Each R_i factor is $+1$, forcing the one unrestricted norm factor to be $+1$ too. There is no missing assumption on the norm modulo four.

Here a is even and $\Delta \equiv 3 \pmod 4$. If f is odd then $S \equiv I^2 \pmod 4$; if f is even then $S \equiv I^2 - 1 \pmod 4$. Thus S is never 2 modulo four. The same is true of D: for even c its residues are $0, 3$, and for odd c they are $0, 1$. Finally $4G \equiv 0 \pmod 4$. All the stated finalizers therefore recover their individual residuals, in particular $S = 0$, before interpreting a rational norm.

Since $a \in 6\mathbb{Z}$, $\Delta = (a+1)(a+3) \geq 3$. In ordinary variants $S = 0$ excludes $Q = 1$, as it would give $\Delta(f^2 - 1) = 1$. In the normalized variant $Q = \Delta^2 I^2$ excludes $Q = 1$ directly. Hence $\delta \neq 0$ at every zero in question.

4.4 Prime separation before guard recovery

The fixed residues give

$$q \equiv X \equiv 2 \pmod 3, \quad a \equiv H \equiv \Delta \equiv 0 \pmod 3, \quad A_\tau \equiv A_m \equiv 1 \pmod 3.$$

To check A_τ, write $A_\tau = 1 + b(b+1)k^2$ with $b = XY^2 \equiv 0$ or 2 modulo three. Also H is nonzero and odd. Under $S = 0$, the square $Q = I^2$ is divisible by three, hence by nine. The normalized Q is already divisible by nine. Thus A_y is a three-adic unit of residue one.

The weighted roots used by the ordinary fourteen-variable norm have squared orders

$$-2v_3(H), \quad 0, \quad 2, \quad 4 + v_3(A_g)$$

for $d_0/H, y, 3\tau, 9p_1$ respectively. The normalized construction inserts $9f$ of squared order four and uses $27p_1$ of squared order at least six. A zero guard root is harmless.

Here is the precise separation argument. In any subset of at least two nonzero roots, the smallest squared order is unique and even. Divide out its power of three and multiply the putative zero sum over all signs of the remaining roots. The product is a polynomial in the rational squares and is a nonzero unit modulo three, a contradiction. Subtracting a Galois conjugate of a rational signed root sum would give exactly such a forbidden signed subsum. Every root is therefore rational. This argument allows negative radicands; their rationality subsequently rules that case out. No positivity guard is assumed beforehand.

Roots of integer squares are integral. For the auxiliary root, $Q = 0$ gives $y = \pm 1$. For $Q > 1$, rational y in $QV^2 - (Q-1)y^2 = 1$ is integral by adjacent-Pell descent. In lowest terms $y = A/m$ gives $m^2 \mid Q - 1$. The transformation

$$V' = (2Q-1)V - 2(Q-1)y, \qquad y' = (2Q-1)y - 2QV$$

preserves the norm and the denominator, and after taking positive signs satisfies $0 < V' < V$ when $V > 1$. The endpoint $|V| = 1$ forces $m = 1$. Thus the case $c = 0$, and consequently $Q = 0$, has not been silently excluded.

4.5 Coprime carriers and exact source restoration

Because $Y = qv$, we have $H \equiv 3 \pmod{q}$. Since $q \equiv 2 \pmod{3}$ and H is odd,

$$\gcd(H, 2q^2) = 1.$$

After root recovery the ordinary fourteen-variable factor is

$$U(n + V) + Mb_0 + He = Md_0 + U(3\tau + y + 9p_1).$$

Reduction modulo M and H gives respectively

$$v = q^2(2v_{\mathrm{old}} + 1), \qquad \gamma = \frac{d_0 - b_0}{H} \in \mathbb{Z}.$$

The normalized case adds $9f$ and replaces $9p_1$ by $27p_1$; the same argument applies. The centered main carrier instead has $2M(d_0 - b_0)$ modulo H, still invertible because H is odd. The variants retaining γ need only the first congruence. In all cases $Y = (2v_{\mathrm{old}} + 1)q^3$ exactly, including its odd quotient. The removed v^2 guard multiplier is unnecessary: $v \neq 0$ follows now from the recovered congruence, not from an assumed inequality.

The integral guard root restores $G = 0$, which implies $d > 0$ and $\eta\zeta > 0$ by the standard joint-positivity identity. For completeness, if both factors were negative their magnitudes $b, e > 0$ would require $(t^2 + b + e + 1)^2 < be$, which is impossible. Thus $c = 0$ is impossible too, since it would give $\eta\zeta = -Y(Y + 1)k^2 \leq 0$.

For the ordinary variants set old $J_3 = 2j$, old scale v_{old} as above, and restore the two centered source coordinates by

$$\tau_{\mathrm{old}} = 2\tau - (2u + k), \qquad y_{\mathrm{gap}} = y - V.$$

The first and auxiliary parent residuals become exactly

$$\tau_{\mathrm{old}}^2 + 2\tau_{\mathrm{old}}(2u + k) + k^2 - 4 = 4(\tau^2 - A_\tau),$$
$$V^2 + (1 - Q)y_{\mathrm{gap}}(2V + y_{\mathrm{gap}}) - 1 = QV^2 - (Q - 1)y^2 - 1.$$

The other three rows are the main norm, S and G. Every remaining parent abbreviation agrees literally. This restores the five-row signed CRT source directly, without inventing a new rank lemma. For normalized thirteen variables instead set

$$r_{\mathrm{old}} = \frac{\Delta(2r + 1) - 1}{2}, \qquad T_{\mathrm{old}} = fT.$$

The first quotient is integral because Δ is odd. Then $I_{\mathrm{old}} = \Delta I$, the parent's V agrees exactly, and its strong row follows from $f^2 = A_f$.

Conversely, the inherited canonical refutation witnesses have odd $J \equiv 1 \pmod{3}$, so $J = 6j + 1$. Put $v = q^2(2v_{\mathrm{old}}+1)$ and retain their guard witnesses. For example the ordinary fourteen-variable construction chooses

$$n = \gamma + 3\tau + y + 9p_1 - V - e/M.$$

The centered fifteen-variable construction chooses

$$n = 6\tau + 2y + 2\gamma - 3t_0 - 2V - e/M.$$

The other formulas omit the coordinates not fused by their norms. All values are integers and make one conjugate factor vanish. The normalized thirteen-variable case uses the same canonical odd strong normalization and fresh auxiliary reconstruction proved in the incoming prime-separation package. Its ordinary calibration boundary is unchanged.

4.6 Universal degree bounds and actual coefficient certificates

Fix the six ports. The exact propagated prefix bounds are

register	H	k	u	A_τ	A_g	Q	V	M	U	S, D
degree	4	5	11	22	28	6	7	2	6	10

in the ordinary case, with $\deg G = 28$. The normalized case instead has $\deg A_f = 14$, $\deg Q = 22$ and $\deg V = 19$. Direct degree propagation through the exact DAGs proves the final upper bounds $378, 172, 98$ for ordinary fourteen, ordinary fifteen and centered sixteen variables. Two cancellation-sensitive cases require more.

Thirteen variables. Pair the two signs of the auxiliary root first. Write

$$Z = UV + s, \qquad s = Un + Mb_0 + H(v - q^2),$$

and, for each of the sixteen sign choices, put

$$B = s \pm 9Uf \pm 3U\tau \pm Md_0 \pm 27Up_1.$$

Assign formal root weights $7, 11, 5, 14$ to f, τ, d_0, p_1. Each squared radicand has degree at most twice its root weight, so reduction in this formal root algebra never increases weighted degree. Thus B has weight at most 20. The paired factor is exactly

$$\delta\{(UV + B)^2 - U^2 C_y/\delta\} = \delta B^2 + 2\delta UVB - U^2(V^2 - 1).$$

The identity uses $C_y = (\delta + 1)V^2 - 1$. Its three terms have bounds $62, 67, 50$. The sixteen-factor product therefore has degree at most $16 \cdot 67 = 1072$, and is exactly the cleared five-root norm. This argument is polynomial algebra, including all exceptional inputs.

Centered fifteen variables. Use the exact centered two-root identity with its scale M replaced by U, and its carrier argument replaced by

$$x = Un + 2Mb_0 + H(v - q^2) \pm 2Md_0.$$

With d_0 of weight five, x has weight at most seven. Define

$$j = \delta\{x^2 + U(6t_0 + 4V)x + 9U^2\ell\} - 4U^2(V^2 - 1),$$
$$h = 12\delta t_0 VU^2.$$

Their weights are at most 30 and 36. Every term of

$$j^2 + 2hj + 576\delta U^4 A_\tau(1 - V^2) + 144\delta^2 U^4 \ell V^2$$

has weight at most 66. The exact identity uses only $t_0^2 = 4A_\tau + \ell$, not a Pell zero. Taking the two main-root conjugates gives norm degree at most 132; the strong and guard finalizers add $10 + 28$, yielding 170.

Independent modular-line certificates modulo 65521 and 65519, using the actual six numeral byte strings, attain every stated bound with a nonzero leading coefficient. They are rigorous lower certificates combined with the preceding universal upper proofs. All literal DAGs pass topology, variable closure, liveness and coefficient hash checks. An independent exact sparse-algebra audit verifies every prefix, carrier, radicand, norm and finalizer, using all 20 349 coefficients of the generic five-root norm; deliberate corruptions are rejected. Signed modular diagnostics additionally include $c = 0$, $v = 0$ and their intersection. These algebraic checks do not establish an unbounded compiler theorem: the named ordinary source calibration remains an explicit mathematical dependency.

5 Twelve integers at degree 2048

A late supplied interpolation theorem permits a further synthesis. The result is a literal polynomial in twelve unrestricted integer variables, of exact degree 2048, evaluated in 233 live binary operations (148 multiplications, 53 additions and 32 subtractions). It improves the supplied twelve-variable 4832/248 degree–cost point. The supplied 16000/223 point remains a cheaper, higher-degree alternative. No fewest-variable or global optimality claim is made.

The new changes are a coprime homogeneous quintic carrier, a guard Pell witness divisible by three, fixed prime separation in place of large size weights, and a doubled auxiliary-root center. The full-residue interpolation theorem and the direct normalized signed-kernel calibration are imported from `ConZFC_Twelve_Integer_Interpolation_2026-10-24`. Its exact proofs, provenance and notices are retained. In particular, this is not a literal restoration of the earlier odd-multiplier source chart, nor a new proof-assistant certificate.

5.1 Complete polynomial specification

The twelve coordinates are

$$j, C, z, h, r, v, w, T, c, n, t, s_0 \in \mathbb{Z}.$$

Use the same complete six fixed source-34/source-71 numerals as before. Define

$$\begin{aligned}
J &= 6j+1, & q &= \beta J+1, & W &= 12w+4,\\
X &= qW, & Y &= qv, & E &= XY,\\
a &= E+Y, & H &= 4a+3, & \Delta &= a^2+H,\\
F &= (K_0+W)C - z(q-1), & & & &\\
R &= (q(q-F)-C)(q^2-1) + (M_C+qM_F)J, & & & &\\
k &= R+1+hE, & \eta &= c-Yk, & \zeta &= k-\eta.
\end{aligned}$$

Put

$$\begin{aligned}
d_g &= (A_{\mathrm{cone}}F)(B_{\mathrm{cone}}C) - (q - A_{\mathrm{cone}}F - B_{\mathrm{cone}}C)^2,\\
P &= d_g\eta\zeta - (t^2 - d_g - \eta\zeta + 1)^2,\\
s &= 3s_0, & D_g &= (3P-4)s^2+1, & I &= src^2,\\
D_f &= 1+\Delta I^2, & Q &= \Delta^2 I^2, & \delta &= Q-1,\\
V &= D_f(c-R+4IT)-c, & C_y &= QV^2-1, & D_a &= \delta C_y,\\
\ell &= EYk, & D_\tau &= 1+\ell(\ell+k), & D_m &= 1+\Delta c^2,\\
b_0 &= X+ac, & M &= 2q^2, & L &= MH,\\
e &= v-q^2, & \widetilde{b} &= Mb_0+He, & u &= LT-\widetilde{b},\\
\widetilde{D} &= M^2 D_m.
\end{aligned}$$

For the fixed intersective quintic

$$\pi(x) = (x^2+31)(x^3+x+1), \qquad \Phi_L(x) = L^5\pi(x/L),$$

the following integer polynomials are its exact quadratic remainder:

$$\begin{aligned}
A &= u^5 + 10u^3\widetilde{D} + 5u\widetilde{D}^2 + 32L^2(u^3+3u\widetilde{D})\\
&\quad + L^3(u^2+\widetilde{D}) + 31L^4u + 31L^5,\\
B &= 5u^4 + 10u^2\widetilde{D} + \widetilde{D}^2 + 32L^2(3u^2+\widetilde{D}) + 2L^3u + 31L^4.
\end{aligned}$$

Thus $A + B\widetilde{d} = \Phi_L(u + \widetilde{d})$ when $\widetilde{d}^2 = \widetilde{D}$. Finally set

$$S = IL^5, \qquad C_a = (2Q-1)V, \qquad Z = S(n + C_a) - A.$$

With $\mathcal{F}_5$ denoting the ordinary 32-sign norm polynomial, the complete equation is

$$\boxed{\mathcal{P}_{12} = \mathcal{F}_5(Z; B^2\widetilde{D}, 4S^2D_a, 36S^2D_f, 324S^2D_\tau, 2916S^2D_g) = 0.}$$

There is no denominator clearing in this norm: all five radicands are integer polynomials. The displayed homogeneous notation is expanded in the actual DAG. No square root, division, variable exponent or additional quantified register occurs in the exported equation.

5.2 Exclude the entire zero multiplier branch first

For all integer inputs q is nonzero, even and 2 modulo three; $X \equiv 2 \pmod 3$ and $a \in 6\mathbb{Z}$. Hence $\Delta = (a+1)(a+3) \geq 3$, and H is nonzero, odd and divisible by three. Also $H \equiv 3 \pmod q$, so $\gcd(H, M) = 1$ and $L \neq 0$. Thus $D_m \geq 1$ and $\widetilde{D} > 0$, including $v = 0$.

The imported elementary quintic lemma says that π has no zero in a real extension of $\mathbb{Q}$ of degree at most two. The quadratic factor has no real zero, while the cubic has degree three and is irreducible by the rational-root test. Consequently

$$A + B\sqrt{\widetilde{D}} \neq 0, \qquad A - B\sqrt{\widetilde{D}} \neq 0.$$

If $I = 0$, then $S = 0$, the last four norm radicands vanish, and the entire polynomial specializes to

$$(A^2 - B^2\widetilde{D})^{16} \neq 0.$$

Every zero therefore has $I, S, s_0, r, c \neq 0$. This is established before dividing a factor relation by S. Since $Q = (\Delta I)^2$, at such a zero $Q > 1$ and $\delta \neq 0$.

5.3 Small fixed weights replace all size separators

Modulo three the new carrier has

$$M \equiv 2, \quad L \equiv 0, \quad b_0 \equiv 2, \quad u \equiv 2, \quad \widetilde{D} \equiv 1.$$

Thus $B \equiv 5 + 10 + 1 \equiv 1 \pmod 3$. Also

$$D_f \equiv D_\tau \equiv D_a \equiv D_g \equiv 1 \pmod 3.$$

For D_a use $Q \equiv 0 \pmod 9$; for D_g use $s = 3s_0$. For D_τ, the factorization $1 + (XY^2)(XY^2+1)k^2$ has $XY^2 \equiv 0$ or 2 modulo three.

Divide a vanishing factor by the already nonzero S. The five weighted roots

$$\frac{BMd}{S}, \quad 2\widetilde{y}, \quad 6f, \quad 18\tau, \quad 54g$$

have squared three-adic orders respectively

$$-2v_3(S), \quad 0, \quad 2, \quad 4, \quad 6,$$

where $d^2 = D_m$, $\widetilde{y}^2 = D_a$, $f^2 = D_f$, $\tau^2 = D_\tau$, $g^2 = D_g$. Each subset of at least two has a unique even minimum. Normalize its smallest root by the corresponding power of three and multiply a putative zero sum over all signs of the other roots. The product is a polynomial in the

rational squared roots and is a nonzero unit modulo three, a contradiction. Subtracting a Galois conjugate of the rational factor relation would give just such a forbidden signed subsum. All roots are therefore rational. The nonzero rational multiplier BM/S causes no loss: d is rational too. Roots of integer radicands are integral. No size estimate, radical positivity assumption or bound on R was used for this extraction. Global monotonicity of the chosen quintic, although true, is not needed by this new proof: $B \equiv 1 \pmod 3$ already supplies its nonvanishing.

Now $D_g = g^2 \geq 0$. Since $s \neq 0$, the alternative $P \leq 0$ would give $D_g = (3P-4)s^2 + 1 \leq -3$. Thus $P > 0$, and the joint-positivity identity gives $d_g > 0$ and $\eta\zeta > 0$. The rational value $y = \widetilde{y}/\delta$ satisfies

$$QV^2 - (Q-1)y^2 = 1.$$

Adjacent-coefficient Pell descent makes y integral.

5.4 One monic carrier restores both congruences

After absorbing the factor signs into the integer roots, the norm relation gives

$$S \mid A + BMd = \Phi_L(Md - \widetilde{b} + LT).$$

In particular L^5 divides this homogeneous monic quintic. For a prime with $p^a \| L$, if its argument had order $b < a$, the monic fifth-power term would have order $5b$, strictly below every other term. That contradicts divisibility by p^{5a}. Therefore

$$L \mid M(d - b_0) - He.$$

Reduction modulo the coprime factors M, H gives

$$M \mid e, \qquad H \mid d - b_0, \qquad v = q^2(2v_{\mathrm{old}} + 1), \qquad \gamma = (d - b_0)/H \in \mathbb{Z}.$$

Thus $Y = (2v_{\mathrm{old}} + 1)q^3$ is recovered with its full odd quotient. Moreover the quintic argument divided by L is exactly

$$T + \gamma - e/M.$$

This proves both congruences without adding a variable or presupposing that v is nonzero.

The late source's direct normalized signed-kernel proof now applies. The physical coordinates satisfy its first, main and normalized strong Pell equations, strict guards and native transport/index equations. Its two auxiliary quotients are restored by

$$o = f(c - R + 4IT),$$
$$j_{\mathrm{aux}} = \Delta s^2 r^2 c^3 (c - R) + 4srcD_f T,$$

so $V = fo - c = cj_{\mathrm{aux}} - R$. The normalized rank argument uses $c^2 \mid I$ and does not require I/c^2 to be odd. Accordingly, we do not assert that the old coordinate $(\Delta sr - 1)/2$ is integral. The inherited signed step-down, residue orientation and native compiler then produce a finite syntactic refutation, with their ordinary theorem status unchanged.

5.5 Completeness and the exceptional guard value

Start with the late source's canonical native witnesses and set $v = q^2(2v_{\mathrm{old}} + 1)$. Choose its nearest-square coordinate t so $P > 0$. A crucial exception must be excluded before forcing $3 \mid s$: if $P = 1$, the guard equation would be $g^2 + s^2 = 1$ and no nonzero multiple of three could serve as s.

Here that exception is impossible. Both cone products and q are even, so $d_g \equiv 0 \pmod 4$. Therefore

$$P \equiv -(t^2 - d_g - \eta\zeta + 1)^2 \equiv 0 \text{ or } 3 \pmod 4.$$

Positive P is at least three. Thus $D = 3P - 4$ is positive and 2 modulo three, hence nonsquare. Take a positive Pell pair $g_1^2 - Ds_1^2 = 1$. Modulo three, $g_1^2 + s_1^2 = 1$, so $3 \mid g_1 s_1$. Squaring the unit gives

$$g = g_1^2 + Ds_1^2, \qquad s = 2g_1 s_1 > 0, \qquad 3 \mid s.$$

Set $s_0 = s/3$. No outer coordinate has changed.

Invoke the imported full-residue interpolation lemma with $N = s$. It supplies a single normalized Pell pair I, f, with $4sc^2 \mid I$, *before* a residue is requested, and then realizes every class of T modulo this same I in

$$V = f^2(c - R + 4IT) - c, \qquad (\Delta I)^2 V^2 - ((\Delta I)^2 - 1)y^2 = 1.$$

Put $r = I/(sc^2)$. The lemma's hypotheses include even $a + 2$, $R \equiv c \equiv 3 \pmod 4$, all supplied by the canonical source. Its proof uses the corrected period $4n_0 f$ modulo f^2 and the Lucas valuation bound $\gcd(4n_0 f, 4I^2) \mid 4I$; mere infinitude of Pell solutions would not suffice.

Choose an intersective residue for $\pi(T + \gamma - e/M)$ modulo I. Interpolation supplies T, y in that class, leaving I, f fixed. Hence

$$A + BMd = L^5 \pi(T + \gamma - e/M)$$

is divisible by $S = IL^5$. With $\widetilde{y} = \delta y$, choose

$$n = \frac{A + BMd}{S} + 2\widetilde{y} + 6f + 18\tau + 54g - (2Q - 1)V.$$

This is integral and makes one norm factor vanish. The witness order is native data, then t, s, s_0, then I, f, r, then the intersective residue and T, y, and finally n. There is no circular residue choice.

5.6 Exact degree 2048

The relevant polynomial degrees are

register	H	Δ	P	I	D_f	Q	V	D_a	D_g	L
degree	4	8	28	4	16	24	21	90	30	6

and

$$\deg A = 35, \quad \deg B = 28, \quad \deg S = 34, \quad \deg C_a = 45, \quad \deg D_\tau = 22, \quad \deg \widetilde{D} = 14.$$

Pair the two auxiliary signs first. Let B_* denote $Sn - A$ plus the four signed non-auxiliary radical terms. Their weights are $35, 42, 45, 49$ for main, strong, first and guard, so B_* has weight at most 49. The paired factor is exactly

$$B_*^2 + 2SC_a B_* + S^2(V^2 + 4Q - 4),$$

because the polynomial identity

$$((2Q - 1)V)^2 - 4(Q - 1)(QV^2 - 1) = V^2 + 4Q - 4$$

holds universally. Its term bounds are $98, 128, 110$. Taking sixteen sign choices proves degree at most $16 \cdot 128 = 2048$. The unique leading paired term is $2SC_a(\pm 54S\sqrt{D_g})$; the eight negative signs cancel. Thus the full leading form is

$$108^{16} S_0^{32} (C_a)_0^{16} (D_g)_0^8 \neq 0.$$

More explicitly, the coefficient of

$$j^{928} w^{288} v^{320} h^{32} s_0^{128} r^{112} c^{224} T^{16}$$

is

$$\boxed{2^{2064} 3^{1400} \beta^{928} > 0.}$$

The exponent sum is 2048. The h^{32} term uniquely selects ha_0 throughout the top part of k^{32}; T^{16} then uniquely selects $4I_0T$ throughout $(4I_0T - R_0)^{16}$. The remaining factors are monomials, so no other fixed coefficient can cancel this term.

Independent actual-coefficient line certificates modulo 65521 and 65519 also attain degree 2048. The complete 233-operation DAG has no dead gates or hidden input leaves. Independent arithmetic checks use a separate binomial quintic remainder and full 32-sign product, including every intersection of the exceptional slices $s_0 = 0, r = 0, c = 0$. These finite checks do not replace the unbounded interpolation or native calibration proofs. The resulting consistency equivalence is an ordinary mathematical theorem composed with those explicitly retained sources, not a proof of consistency or a completed object-language ZFC derivation.

6 A sharp limit for final coordinate centering

The ten-integer equations now have the following literal circuit counts. The comparison uses the current pinned short-guard family, rather than an older incoming package. These changes preserve the corresponding polynomials exactly; they do not improve the variable counts or degrees.

Variant	Prior operations	New operations	Exact degree
Polar	471	467	1,245,324
Linear	475	470	1,179,788
Cubic	478	473	1,114,252
Six term	512	507	1,080,888

Every live binary addition, subtraction and multiplication costs one, including squares and scalar products. Fixed integer leaves and reuse cost zero. The full graphs, actual coefficients and independent checks are in `norm/`; no intermediate register adds an unknown.

The more substantive result is a limitation of this entire centering route. With the short-guard prefix fixed, no polynomial translation of the final coordinate alone can produce degree below 1,080,888. This includes arbitrary real polynomial translations, not only the integral corrections previously considered. It is not a lower bound for different guards, norm carriers, source encodings, or changes of other coordinates.

6.1 The fixed norm and its leading forms

Retain the ten coordinates $(t, g_0, c_0, k_0, l, w, x, y, z, m)$ and the source registers from the supplied short-guard parent. In particular,

$$\Delta = g(\mathrm{cap} - g)K^2 - \mathrm{cap}^2(C - KL)^2, \qquad a_0 = r + (9\Delta - 3)(rz)^2, \qquad a_1 = DFI.$$

Set $Z = \Sigma m - \Theta$. The unpolarized final norm is

$$N = (Z^2 - \Sigma^2(a_0 + a_1))^2 - 4\Sigma^4 a_0 a_1.$$

For a nonzero polynomial P, write P_{top} for its highest homogeneous part. The relevant degrees are

P	g	cap	K	L	r	a_0	a_1
$\deg P$	2	233	28548	4220	114189	294382	458818.

The first term of Δ has degree 57331, while its negative square term has degree 66002. Therefore, with

$$U = 3\,\mathrm{cap}_{\mathrm{top}} K_{\mathrm{top}} L_{\mathrm{top}} r_{\mathrm{top}} z, \qquad V = (2AEGH)_{\mathrm{top}},$$

we have

$$(a_0)_{\mathrm{top}} = -U^2, \qquad (a_1)_{\mathrm{top}} = V^2.$$

The second identity also follows from the exact defect identity: $\deg(a_1 - (2AEGH)^2) = 426050 < 458818$. Both U and V are nonzero. Since $\deg \Sigma = 81922$, the fixed cross term has degree

$$d = 4(81922) + 294382 + 458818 = 1080888$$

and highest homogeneous part

$$(-4\Sigma^4 a_0 a_1)_{\mathrm{top}} = (2\Sigma_{\mathrm{top}}^2 UV)^2.$$

This is a nonzero square, not merely a coefficient whose sign was sampled.

6.2 The lower bound for every polynomial translation

Let R be any real polynomial in the first nine coordinates, and define

$$B_R = [\Sigma(m + R) - \Theta]^2 - \Sigma^2(a_0 + a_1).$$

The translated norm is $B_R^2 - 4\Sigma^4 a_0 a_1$. If $2\deg B_R > d$, its degree is $2\deg B_R > d$. If $2\deg B_R < d$, its degree is d. At equality, its degree-d part is

$$(B_{R,\mathrm{top}})^2 + (2\Sigma_{\mathrm{top}}^2 UV)^2.$$

A sum of squares of real polynomials can vanish identically only when both summands vanish identically. The second is nonzero, so cancellation is impossible. Thus every such translated norm has degree at least d. The six-term shift attains d, proving the restricted-family optimum. The old odd-degree argument does not apply after the short-guard change; this positive-leading-square argument replaces it.

Further degree progress in this ten-variable family must therefore change more than the final polynomial translation: for example the guard, the norm carrier, the prefix, or another coordinate. This conclusion does not rule out any of those routes.

6.3 The exact arithmetic savings

Write $t_c = t_{c,0} - 2$ in the unchanged source. Reuse

$$X_+ = n_n(s_c + t_{c,0} - 1) - s_c, \quad X = X_+ - 1, \quad K = X_+ + \mathrm{twice_k}(P_{1/2} - 1).$$

This removes two unnecessary constant adjustments. Since $F = F_{-1} + 1$, use $G = hF_{-1} - F$ instead of $(h-1)F_{-1} - 1$. Share $S^3 = SS^2$ and $S^5 = S^2S^3$ between Σ and Θ. For the linear and cubic shifts, share GH by writing $I = (GH)^2 - H^2 + 1$.

For the six-term shift put $u = C^2$, $h = A/2$ and $E_0 = u\xi$. Its compact factorization uses

$$u^7\xi^4 = u^3E_0^4, \qquad u\xi^2 = E_0\xi.$$

These save two multiplications without changing the existing shift. Together the exact identities account for the counts in the table.

6.4 Verification and mathematical boundary

The independent checker imports neither the producer nor its circuit helper. It checks exact local polynomial equalities for every retained source register and the complete norm, all actual coefficient payloads and recipes, topology, backward liveness, operation counts, the expanded radical defects, and nonzero actual-coefficient degree certificates. It also performs 360 signed modular native replays per variant and rejects four final-sign mutations. Normal and optimized Python runs agree.

The new algebra preserves the inherited ordinary mathematical equivalence to finite ZFC refutability. Its compiler, Pell and interpolation interfaces remain explicit dependencies. The result is not a consistency proof or a new end-to-end proof-assistant certificate. No new low-degree integer chart was established here: the incoming 23/10 and 20/18 points remain prior work rather than contributions of this iteration.

7 What the checks establish

The release separates three different levels of evidence.

1. **Literal finite algebra.** Independent programs re-expand the quadratic rows of the selected direct quartics and compare every collected integer coefficient. They verify every retained variable, affine restoration, source binding, Pell macro, radius certificate, and the integer sign closure. Fixed norm payloads are authenticated and their arithmetic graphs checked for topology, liveness and operation counts.
2. **Universal mathematical reductions.** The carrier bootstrap, beta reconstruction, prime-lift CRT, forward Pell witnesses, parametric elimination, unit-product finalizers, root separation and degree upper bounds are proofs about arbitrary inputs. Finite samples are not offered as substitutes for them. Exact modular leading coefficients, once an upper bound is proved, are valid certificates that the degree is attained.
3. **Logical calibration.** The direct route retains the named numerical C222 and ZF/ZFC refutation interfaces. The low-variable route retains the native compiler and K215 interfaces. Their ordinary source proofs are included or identified at their explicitly recorded source pins. A new complete Lean or Rocq derivation is not claimed. The package does not search for, or decide the existence of, a zero of a consistency polynomial.

The exact audit programs use explicit failures rather than removable Python assertions. The release driver invokes the selected suites both normally and with Python optimization enabled. It also regenerates the selected equations in temporary directories and compares their bytes, not merely their numerical summaries. Source and proof hashes, payload hashes, check outputs and timing receipts are recorded separately. Checks that write receipts are run on temporary copies, so verification does not silently invalidate the release manifest.

The archive README lists the exact suite inventory and supported command. The runtime receipt records actual counts and elapsed times of the replay; it is not a promise about speed on another machine. Python 3 and SymPy are required for the complete suite. The direct, architecture and radical checks themselves use standard-library Python. No network access,

repository checkout, Lean installation or earlier round's ZIP is required. Rebuilding this PDF additionally requires a conventional LaTeX setup. The PDF is delivered already rendered, so it is not needed to check the mathematics encoded in the JSON files.

7.1 Immediate dependency map

The root comparison pin and baseline recount are recorded under `review/`. The named direct calibration interfaces are `c222_iff_zf_refutation` and `zf_refutation_iff_zfc_refutation`; the endpoint uses the selected `ZFCax_s` proof predicate. The direct packet retains the exact constructor source, Horn branch inventory, C222 endpoint, constructible-choice bridge, Pell/power/factorial and bounded-quantifier interfaces under `direct/sources/`. The official Mathlib indexed-Pell completeness source is fixed independently by its own commit, Git blob and SHA256. Its source notice and Apache license are preserved.

The radical packet retains the five-row seventeen-integer parent and its CRT and prime-separation proofs. Its six coefficient payloads are not interchangeable with a different native tuple. In particular, the already-shifted M_F must not be shifted again. The ten-variable packet retains its twelve literal signed integer leaves, their source recipes, the exact short-guard parent graphs and the parent proof. Included source inventories distinguish a local dependency from a historical path mentioned in inherited prose. Historical source references are not runtime dependencies.

8 What to pursue next

The direct quartic is now substantially smaller without changing the logical endpoint. The remaining sign seeds and the repeated arithmetic macros are specific targets, but further compression must preserve simultaneous completeness. Merely orienting a congruence or sharing a bound because it looks harmless is insufficient.

1. **Extend the sign certificate before changing the integer encoding.** The current 414 guarded signs, 80 normalized pivots and 35 inferred signs establish all 529 nonquotient semantic signs. A new universally valid implication that removes a seed saves three unrestricted integer coordinates. It must be checked against the literal residual and the dependency order; signed quotient cones must remain outside the natural arithmetic semantics.
2. **Seek genuinely shared power certificates.** There are still 40 positive-power calls. Any proposed shared Pell parameter must be chosen after all its finite input requirements and must satisfy the simultaneous index and final-norm conditions. A common large numerical bound alone does not prove that those Pell witnesses can be shared. A proof of this joint existence theorem would be more valuable than another local circuit reassociation.
3. **Change the short guard or prefix for ten variables.** The proved 1080888 lower bound closes the route consisting only of final polynomial translations. Further degree improvement in that family requires changing another structural element. The restriction is sharp and useful, but says nothing about arbitrary ten-variable equations.
4. **Treat another eliminated guard coordinate as a new theorem.** The tempting replacement of $1 + p_1^2 + p_2^2 + p_3(p_3 + 1)$ by $1 + p_1^2 + p_3(p_3 + 1)$ is not uniformly valid. For $d = 1$ and $\eta\zeta = 9$, the positive joint expression is $9 - (t^2 - 9)^2$. Its only positive values occur at $t = \pm 3$ and equal 9. But 8 is not a square plus $z(z + 1)$: the only nonnegative consecutive products at most 8 are $0, 2, 6$, leaving $8, 6, 2$, none a square. A twelve-variable proposal based on this shortcut therefore needs an additional canonical-witness tuning argument, not an omitted exception. This failed shortcut is not the interpolation construction of Section 5.
5. **Formalize the smallest complete chain.** A useful next ProveIt milestone is a checked translation for the direct 1859-variable quartic: first the fixed Horn factoring and degree-

bootstrap carrier, then the short-beta and majorant transformations, then the parametric Pell compiler. The supplied finite checkers identify concrete interfaces but do not themselves create those universal formal theorems.

9 Reading and reproducing the archive

The archive is self-contained at the level claimed here. Its root contains this report's editable TeX and rendered PDF, a README, the release verifier, a SHA256 manifest, and a recorded full offline replay. The component directories contain only the selected equations and their immediate proof, coefficient and verification dependencies. Earlier complete incoming archives and redundant development variants are not bundled.

Start with the README, then run `python verify.py`. This validates the manifest and executes the full offline check and deterministic rebuild suite. `python -O verify.py` provides the corresponding optimized-driver replay. Outputs made while checking go to an independent temporary directory; the release itself is left unchanged. To read a particular equation, use the domain and variant name in the tables above, not a mixed-domain intermediate template.

The polynomial payloads are the authoritative exact artifacts. The human-readable proofs explain why they have the required existential meaning. Neither a successful run nor a small variable count should be read as a proof that the polynomial has no zero, as global optimality, or as a substitute for the fixed syntactic definition of consistency.